Output Input Input

0 1 2 3

Output

new_all_borrowed

0

0 1 2

TailLoop

0 1

borrow_array<20,...

Const(0)
0

0
LoadConst
0
int<6>

0
Call
0

0

UnpackTuple
0 1 2

Tuple(int<6>,...)

0 1 2

MakeTuple
0

int<6>

int<6>
int<6>

Tuple(int<6>,...)

Input
0 1 2

0

UnpackTuple
0 1 2

Tuple(int<6>,...)

0 1 2
Conditional
0 1 2

borrow_array<20,...
int<6>

0 1 2

Output

0 1 2

MakeTuple
0

int<6>

int<6> int<6>

0 1
Call
0

Tuple(int<6>,...)

Option(Tuple...)

Input
0 1

0 1 2

Output

borrow_array<20,...
int<6>

Right

0

Either(Tuple...)

Case

Input
0 1 2

0

UnpackTuple
0 1

Tuple(int<6>,...)

0
itousize
0

int<6>

0 1 2
return
0

borrow_array<20,...

0 1
iadd
0

int<6>

0 1 2

Output

0

UnpackTuple
0 1 2

Tuple(int<6>,...)

0 1

MakeTuple
0

int<6>

0 1 2

MakeTuple
0

int<6>

int<6>
int<6>

Tuple(int<6>,...)

0

UnpackTuple
0 1

Tuple(int<6>,...)

0
Left
0

Tuple(int<6>,...)

Either(Tuple...)

QAlloc
0

Qubit
USize

borrow_array<20,...

Const(1)
0

0
LoadConst
0
int<6>

int<6>

Case

Either(Tuple...)
borrow_array<20,...

int<6>

0 1
borrow
0 1

borrow_array<20,...

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(4)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(5)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(6)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(7)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(8)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(9)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(10)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(11)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(12)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(13)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(14)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(15)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(16)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(17)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(18)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(19)
0

0
LoadConst
0

0
itousize
0

int<6>
0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

borrow_array<20,...

MakeTuple
0

MakeTuple
0

QAlloc
0

0
H
0

Qubit

Qubit

MakeTuple
0

QAlloc
0
Qubit

Tag(0)
0
Unit

DataflowBlock
0

0
DataflowBlock
0 1

0
ExitBlock

Input
0 1 2

0 1 2

Output

borrow_array<20,...
Qubit

0
MeasureFree
0

Qubit

0
read
0

bool

Bool

0
DataflowBlock
0

0
DataflowBlock
0

Input
0 1

0 1 2
Call
0 1

borrow_array<20,...
Qubit

0 1 2 3

Output

0 1 2
Call
0 1

borrow_array<20,...
Qubit

MakeTuple
0

0 1 2 3
Call
0 1 2

borrow_array<20,...

Qubit

MakeTuple
0

QAlloc
0
Qubit

Qubit
borrow_array<20,...

Qubit

MakeTuple
0

Tag(0)
0
Unit

Input
0 1

0 1
Call
0

borrow_array<20,...

0
QFree

Qubit

0

Output

0
clone
0 1

borrow_array<20,...

0 1
scan
0

borrow_array<20,...
0

drop

borrow_array<20,...

0
LoadFunc
0
bool -> Bool

0

to_array
0

borrow_array<20,...

0

result_array_bool

array<20,...

MakeTuple
0

MakeTuple
0

Tag(0)
0

Unit

CFG

[FuncDefn(__main__...)]
unitary: 0

0

Input

0

Output

Input

0 1

Output

Const(0)
0

0
LoadConst
0

0 1 2

MakeTuple
0

int<6>

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

int<6>

Const(1)
0

0
LoadConst
0
int<6>

Tuple(int<6>,...)

Tag(0)
0

Unit

DataflowBlock
0

0
ExitBlock

CFG
0

Tuple(int<6>,...)

FuncDefn(guppylang...)
unitary: 0

0

Input
0

0
CFG
0

Tuple(int<6>,...)

0

Output

Input
0

0

UnpackTuple
0 1 2

Tuple(int<6>,...)

0 1 2 3

Output

int<6>

int<6>

int<6>

0 1

ige_s

0

int<6>

Const(0)
0

0
LoadConst
0
int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

DataflowBlock
0 1

0
DataflowBlock
0

0
DataflowBlock
0

0
ExitBlock

Input
0 1 2

0 1 2 3 4

Output

int<6>

int<6> int<6>

0 1

ige_s

0

int<6>
int<6>

0

make_opaque

0

Bool

bool

Tag(0)
0

Unit

0
DataflowBlock
0 1

Input
0 1 2

0 1 2 3 4

Output

int<6>

int<6> int<6>

0 1

ile_s

0

int<6>
int<6>

0

make_opaque

0

Bool

bool

Tag(0)
0

Unit

Input
0 1 2 3

0
read
0

bool

0 1 2 3
Conditional
0

int<6>
int<6>

int<6>

0

Output

Bool

Input
0 1 2

0 1 2
Left
0

int<6>
int<6>

int<6>

0

Output

Either(...)

Case

Input
0 1 2

0

Output

Right

0

Either(...)

Case

Either(...)

0
DataflowBlock
0

0
DataflowBlock
0

Input

0 1

Output

Left
0

Option(Tuple...)

Tag(0)
0

Unit

Input
0 1 2

0 1
iadd
0

int<6>
int<6>

0 1 2

MakeTuple
0

int<6>
int<6>

0 1

MakeTuple
0

int<6>

0 1

Output

int<6>

Tuple(int<6>,...)

0
Some
0

Tuple(int<6>,...)

Option(Tuple...)

Tag(0)
0
Unit

Option(Tuple...)

FuncDefn(guppylang...)
unitary: 0

0

Input
0 1

0 1
CFG
0 1

borrow_array<20,...
Qubit

0 1

Output

Input
0 1

0 1
borrow
0 1

borrow_array<20,...

0 1 2
Call
0 1

Qubit

0 1 2

Output

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(4)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(7)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(9)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(10)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(13)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(17)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(19)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Qubit

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(4)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(7)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(9)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(10)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(13)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(17)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(19)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Tag(0)
0
Unit

DataflowBlock
0

0
ExitBlock

borrow_array<20,... Qubit

FuncDefn(__main__...)
unitary: 0

0

0 1 2
Call
0 1

0 1 2
Call
0 1

Input
0 1

0 1
CFG
0 1

borrow_array<20,...
Qubit

0 1

Output

Input
0 1

0 1
borrow
0 1

borrow_array<20,...

0
Z
0

Qubit

0 1 2

Output

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>
USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit
USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(4)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(5)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(6)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(7)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(8)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(9)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(10)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(11)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(12)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(13)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(14)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(15)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(16)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(17)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(18)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(19)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(4)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(5)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(6)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(7)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(8)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(9)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(10)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(11)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(12)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(13)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(14)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(15)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(16)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(17)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(18)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(19)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1 2
Call
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Qubit

MakeTuple
0

0
Z
0

Qubit

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

Qubit

MakeTuple
0

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

USize int<6>

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(4)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(5)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(6)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(7)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(8)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(9)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(10)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(11)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(12)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(13)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(14)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(15)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(16)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(17)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(18)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(19)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
X
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(4)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(5)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(6)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(7)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(8)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(9)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(10)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(11)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(12)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(13)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(14)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(15)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(16)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(17)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(18)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit

USize

0 1
borrow
0 1

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Const(19)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0
H
0

Qubit

0 1 2
return
0

borrow_array<20,...

Qubit
USize

borrow_array<20,...

MakeTuple
0

MakeTuple
0

Tag(0)
0
Unit

DataflowBlock
0

0
ExitBlock

borrow_array<20,...

Qubit

FuncDefn(__main__...)
unitary: 0

0

Input
0 1 2

0 1 2
CFG
0 1 2

borrow_array<20,...
Qubit Qubit

0 1 2

Output

Input
0 1 2
borrow_array<20,...
Qubit

0 1
CZ
0 1

Qubit

0 1 2 3

Output

Qubit

borrow_array<20,...

MakeTuple
0

0
V
0

Qubit

0 1 2
CRz
0 1

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Const(0.1)
0

0
LoadConst
0

0

MakeTuple
0

float64

0

UnpackTuple
0

Tuple(float64)

0

from_halfturns_unchecked

0

float64

rotation

0

Vdg
0

Qubit Qubit

MakeTuple
0

Qubit

MakeTuple
0

borrow_array<20,...

Qubit

MakeTuple
0

Tag(0)
0

Unit

DataflowBlock
0

0
ExitBlock

borrow_array<20,... Qubit

Qubit

FuncDefn(__main__...)
unitary: 0

0

Input
0

0
CFG
0

borrow_array<$0,...

0

Output

Input
0

0 1
Call
0

borrow_array<$0,...

0 1

Output

new_all_borrowed

0

0 1 2

TailLoop

0 1

borrow_array<$0,...

Const(0)
0

0
LoadConst
0

int<6>

0

UnpackTuple
0 1

Tuple(borrow_array<$0,...)

0 1

MakeTuple
0

borrow_array<$0,... int<6>

Tuple(borrow_array<$0,...)

Input
0 1 2

0

UnpackTuple
0 1

Tuple(borrow_array<$0,...)

0 1 2
Conditional
0 1 2

borrow_array<$0,...
int<6>

0 1 2

Output

0 1

MakeTuple
0

borrow_array<$0,... int<6>

0 1
Call
0

Tuple(borrow_array<$0,...)

Option(Tuple...)

Input
0 1

0 1 2

Output

borrow_array<$0,...

int<6>

Right

0
Either(Tuple...)

Case

Input
0 1 2

0

UnpackTuple
0 1

Tuple(Qubit,...)

0
itousize
0

int<6>

0 1 2
return
0

borrow_array<$0,...

0 1
iadd
0

int<6>

0 1 2

Output

0

UnpackTuple
0 1

Tuple(borrow_array<$0,...)

0 1

MakeTuple
0

Qubit
0 1

MakeTuple
0

borrow_array<$0,... int<6>

Tuple(borrow_array<$0,...)

0

UnpackTuple
0 1

Tuple(Qubit,...)

0
Left
0

Tuple(borrow_array<$0,...)

0
MeasureFree
0

Qubit

Either(Tuple...)

bool

USize

borrow_array<$0,...

Const(1)
0

0
LoadConst
0
int<6>

int<6>

Case

Either(Tuple...)
borrow_array<$0,...

int<6>

borrow_array<$0,...

Tag(0)
0

Unit

DataflowBlock
0

0
ExitBlock

borrow_array<$0,...

FuncDefn(guppylang...)
unitary: 0

0

Input
0

0
read
0

bool

0

Output

Bool

FuncDefn(array...)
0

Input
0

0
CFG
0

borrow_array<$1,...

0

Output

Input
0

0 1

MakeTuple
0

borrow_array<$1,...

0 1

Output

Const(0)
0

0
LoadConst
0
int<6>

Tuple(borrow_array<$1,...)

Tag(0)
0

Unit

DataflowBlock
0

0
ExitBlock

Tuple(borrow_array<$1,...)

FuncDefn(guppylang...)
unitary: 0

0

Input
0

0
CFG
0

Tuple(borrow_array<$1,...)

0

Output

Input
0

0

UnpackTuple
0 1

Tuple(borrow_array<$1,...)

0 1

Output

borrow_array<$1,...

0 1

ilt_s

0

int<6>

0 1
Conditional
0

int<6>

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

Input
0

0

Output

Left
0
Option(int<6>)

Case

Input
0

0
Some
0

int<6>

0

Output

Option(int<6>)

Case

Option(int<6>)

DataflowBlock
0 1

0
DataflowBlock
0

0
DataflowBlock
0

0
ExitBlock

Input
0 1

0
itousize
0

int<6>

0 1
borrow
0 1

borrow_array<$1,...

0 1
iadd
0

int<6>

0 1

Output

USize

0 1

MakeTuple
0

borrow_array<$1,...

0 1

MakeTuple
0

$0

Const(1)
0

0
LoadConst
0
int<6>

int<6>

Tuple(borrow_array<$1,...)

0
Some
0

Tuple($0,...)

Option(Tuple...)

Tag(0)
0
Unit

Input
0

0

discard_all_borrowed

borrow_array<$1,...

0 1

Output

MakeTuple
0

Left
0
Option(Tuple...)

Tag(0)
0

Unit

Option(Tuple...)

FuncDefn(guppylang...)
unitary: 0

0

Input
0 1

0 1
CFG
0 1

borrow_array<$0,... Qubit

0 1

Output

Input
0 1

0 1 2

Output

borrow_array<$0,...
Qubit

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

0 1
Call
0 1

int<6>

0 1

MakeTuple
0

int<6> int<6>

0

UnpackTuple
0 1

Tuple(int<6>,...)

0 1 2
Conditional
0

int<6>
int<6>

Const(0)
0

0
LoadConst
0

0 1

ieq

0

int<6>

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

Input
0 1

0 1
Left
0

int<6>
int<6>

0

Output

Either(...)

Case

Input
0 1

0

Output

Right

0

Either(...)

Case

Either(...)

DataflowBlock
0 1

0
DataflowBlock
0

0
DataflowBlock
0 1

0
ExitBlock

Input
0 1

0 1 2

Output

borrow_array<$0,...

0
X
0

Qubit

Qubit

MakeTuple
0

Tag(0)
0

Unit

0
DataflowBlock
0

Input
0 1 2 3

0 1 2

Output

borrow_array<$0,...
Qubit

0 1 2
Conditional
0

int<6>

int<6>

Const(1)
0

0
LoadConst
0

0 1

ieq

0

int<6>

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

Input
0 1

0 1
Left
0

int<6>
int<6>

0

Output

Either(...)

Case

Input
0 1

0

Output

Right

0

Either(...)

Case

Either(...)

0
DataflowBlock
0

0
DataflowBlock
0 1

Input
0 1

0 1
borrow
0 1

borrow_array<$0,...

0 1
CX
0 1

Qubit

0 1 2

Output

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

Qubit

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

Tag(0)
0
Unit

0
DataflowBlock
0

Input
0 1 2 3

0 1 2

Output

borrow_array<$0,...
Qubit

0 1 2
Conditional
0

int<6>

int<6>

Const(2)
0

0
LoadConst
0

0 1

ieq

0

int<6>

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

Input
0 1

0 1
Left
0

int<6>
int<6>

0

Output

Either(...)

Case

Input
0 1

0

Output

Right

0

Either(...)

Case

Either(...)

0
DataflowBlock
0

0
DataflowBlock
0 1

Input
0 1

0 1
borrow
0 1

borrow_array<$0,...

0 1 2
Toffoli
0 1 2

Qubit

0 1 2

Output

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>
USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>
USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

Qubit

0 1 2
return
0

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

Tag(0)
0

Unit

0
DataflowBlock
0

Input
0 1 2 3

0 1 2

Output

borrow_array<$0,...
Qubit

0 1 2
Conditional
0

int<6>
int<6>

Const(3)
0

0
LoadConst
0

0 1

ieq

0

int<6>

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

Input
0 1

0 1
Left
0

int<6>
int<6>

0

Output

Either(...)

Case

Input
0 1

0

Output

Right

0

Either(...)

Case

Either(...)

0
DataflowBlock
0

0
DataflowBlock
0 1

Input
0 1

0 1
borrow
0 1

borrow_array<$0,...

0 1 2 3 4 5
Call
0 1 2 3 4

Qubit

0 1 2

Output

QAlloc
0

Qubit

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

Qubit

0 1 2
return
0

Qubit

0 1 2
return
0

Qubit

0
QFree

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

MakeTuple
0

Tag(0)
0
Unit

0
DataflowBlock
0

Input
0 1 2 3

0 1 2

Output

borrow_array<$0,...
Qubit

0 1 2
Conditional
0

int<6>
int<6>

Const(4)
0

0
LoadConst
0

0 1

ieq

0

int<6>

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

Input
0 1

0 1
Left
0

int<6>
int<6>

0

Output

Either(...)

Case

Input
0 1

0

Output

Right

0

Either(...)

Case

Either(...)

0
DataflowBlock
0

0
DataflowBlock
0

Input
0 1

0 1
borrow
0 1

borrow_array<$0,...

0 1 2 3 4 5 6
Call
0 1 2 3 4 5

Qubit

0 1 2

Output

QAlloc
0
Qubit

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(3)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

Qubit

0 1 2
return
0

Qubit

0 1 2
return
0

Qubit

0 1 2
return
0

Qubit

0
QFree

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

MakeTuple
0

Tag(0)
0

Unit

0
DataflowBlock
0

Input
0 1 2 3

0 1
borrow
0 1

borrow_array<$0,...

0 1 2 3 4 5 6 7
Call
0 1 2

Qubit
int<6>

int<6>

0 1 2

Output

QAlloc
0

0 1 2 3 4
Call
0 1 2 3

Qubit

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

0 1 2
return
0

Qubit

0 1 2
return
0

Qubit

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

Const(3)
0

0
LoadConst
0
int<6>

Const(0)
0

0
LoadConst
0

int<6>

Qubit

0 1
borrow
0 1

borrow_array<$0,...

0 1 2 3 4
Call
0 1 2 3

Qubit

MakeTuple
0

Const(0)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(1)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(2)
0

0
LoadConst
0

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

0 1 2
return
0

Qubit

0 1 2
return
0

Qubit

0
QFree

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

MakeTuple
0

Tag(0)
0

Unit

Input
0 1

0 1 2

Output

borrow_array<$0,...

Qubit

Tag(0)
0
Unit

Input
0 1

0 1 2

Output

borrow_array<$0,...

Qubit

Tag(0)
0
Unit

Input
0 1

0 1 2

Output

borrow_array<$0,...

Qubit

Tag(0)
0

Unit

Input
0 1

0 1 2

Output

borrow_array<$0,... Qubit

Tag(0)
0

Unit

Input
0 1

0 1 2

Output

borrow_array<$0,...

Qubit

Tag(0)
0

Unit

borrow_array<$0,...
Qubit

FuncDefn(__main__.cnx)
unitary: 0

0

Input
0

0 1

imod_s

0

int<6>

0 1
isub
0

int<6>

0 1
isub
0

int<6>

0 1

Output

Const(2)
0

0
LoadConst
0
int<6>

0 1

MakeTuple
0

int<6>

Const(2)
0

0
LoadConst
0
int<6>

0 1

idiv_s

0

int<6>

Const(2)
0

0
LoadConst
0
int<6>

0 1
iadd
0

int<6>

Const(2)
0

0
LoadConst
0
int<6>

0 1

imod_s

0

int<6>

Const(2)
0

0
LoadConst
0
int<6>

int<6>

int<6>

0

UnpackTuple
0 1

Tuple(int<6>,...)

int<6>int<6>

FuncDefn(_get_variables_cnx)

0

Input
0 1 2 3 4

0 1 2 3 4
CFG
0 1 2 3 4

Qubit
Qubit

Qubit
Qubit

Qubit

0 1 2 3 4

Output

Input
0 1 2 3 4

0 1 2 3
Call
0 1 2

Qubit
Qubit

Qubit

0 1 2
Toffoli
0 1 2

Qubit
Qubit

0 1 2 3 4 5

Output

Qubit

0 1 2 3
Call
0 1 2

Qubit
Qubit

MakeTuple
0

Qubit
Qubit

Qubit

MakeTuple
0

Qubit
Qubit

Qubit

MakeTuple
0

Tag(0)
0
Unit

DataflowBlock
0

0
ExitBlock

Qubit
Qubit

Qubit
Qubit

Qubit

FuncDefn(__main__._tof4)
unitary: 0

0

0 1 2 3 4 5
Call
0 1 2 3 4

Input
0 1 2 3 4 5

0 1 2 3 4 5
CFG
0 1 2 3 4 5

Qubit Qubit

Qubit Qubit
Qubit

Qubit

0 1 2 3 4 5

Output

Input
0 1 2 3 4 5

0 1 2 3 4
Call
0 1 2 3

Qubit
Qubit

Qubit Qubit

0 1 2
Toffoli
0 1 2

Qubit

Qubit

0 1 2 3 4 5 6

Output

Qubit

0 1 2 3 4
Call
0 1 2 3

Qubit
Qubit

Qubit

MakeTuple
0

Qubit
Qubit

Qubit

MakeTuple
0

Qubit
Qubit

Qubit Qubit

MakeTuple
0

Tag(0)
0
Unit

DataflowBlock
0

0
ExitBlock

Qubit
Qubit

Qubit
Qubit

Qubit
Qubit

FuncDefn(__main__._tof5)
unitary: 0

0

0 1 2 3 4 5 6
Call
0 1 2 3 4 5

Input
0 1 2 3

0 1 2 3
CFG
0 1 2 3

Qubit
Qubit

Qubit
Qubit

0 1 2 3

Output

Input
0 1 2 3

0
H
0

Qubit

0 1
CX
0 1

Qubit

0 1
CX
0 1

Qubit

0 1
CX
0 1

Qubit

0 1 2 3 4

Output

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

0

Tdg
0

Qubit

0 1
CX
0 1

Qubit

MakeTuple
0

0
H
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

0
T
0

Qubit

0 1
CX
0 1

Qubit

MakeTuple
0

Qubit

MakeTuple
0

0

Tdg
0

Qubit

0 1
CX
0 1

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0

Tdg
0

Qubit

MakeTuple
0

0
H
0

Qubit

MakeTuple
0

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0

Tdg
0

Qubit

MakeTuple
0

0
H
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Tag(0)
0
Unit

DataflowBlock
0

0
ExitBlock

Qubit
Qubit

Qubit Qubit

FuncDefn(__main__...)
unitary: 0

0

0 1 2 3 4
Call
0 1 2 3

Input
0 1 2 3 4 5 6

0 1 2 3 4 5 6
CFG
0 1 2

borrow_array<$0,...
Qubit

Qubit
int<6>

int<6>
int<6>

int<6>

0 1 2

Output

Input
0 1 2 3 4 5 6

0 1 2 3 4

Output

Qubit

borrow_array<$0,...

Qubit

0 1
isub
0

int<6>

0 1

ieq

0

int<6>

0 1 2 3 4
Conditional
0

int<6>
int<6>
int<6> int<6>

QAlloc
0

Qubit

Const(2)
0

0
LoadConst
0
int<6>

int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

Input
0 1 2 3

0 1 2 3
Left
0

int<6>
int<6>

int<6>
int<6>

0

Output

Either(...)

Case

Input
0 1 2 3

0

Right

0

int<6>

0

Output

Either(...)

Case

Either(...)

DataflowBlock
0 1

0
DataflowBlock
0 1

0
DataflowBlock
0

0
ExitBlock

Input
0 1 2 3 4

0 1 2 3 4

Output

Qubit
borrow_array<$0,...

QubitQubit

0 1

ieq

0

int<6>

Const(1)
0

0
LoadConst
0
int<6>

0

make_opaque

0

Bool

0
read
0

bool

Bool

0
DataflowBlock
0

0
DataflowBlock
0

Input
0 1 2 3 4 5 6 7

0
itousize
0

int<6>

0 1
borrow
0 1

borrow_array<$0,...

0 1
iadd
0

int<6> Qubit

Qubit

0
itousize
0

int<6>

0 1
iadd
0

int<6>
0 1
iadd
0

int<6>

0 1 2 3 4 5 6 7
Call
0 1 2

Qubit

int<6>

int<6>
0
itousize
0

int<6>

0 1
iadd
0

int<6>

0
itousize
0

int<6>

0 1 2 3 4

Output

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(1)
0

0
LoadConst
0
int<6>

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

0 1 2
return
0

Qubit

Qubit

0 1 2 3 4
Call
0 1 2 3

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

Const(2)
0

0
LoadConst
0
int<6>

int<6>

Const(1)
0

0
LoadConst
0
int<6>

int<6>

Qubit

0 1
borrow
0 1

borrow_array<$0,...

Qubit

MakeTuple
0

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

Const(1)
0

0
LoadConst
0
int<6>

0
itousize
0

int<6>

0
itousize
0

int<6>
USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit
Qubit

Qubit

0 1 2
return
0

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

Tag(0)
0
Unit

0
DataflowBlock
0

Input
0 1 2 3

0 1
borrow
0 1

borrow_array<$0,...

Qubit Qubit
Qubit

0 1 2 3 4

Output

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

0 1
isub
0

int<6>

Const(2)
0

0
LoadConst
0
int<6>

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

0 1
isub
0

int<6>

Const(1)
0

0
LoadConst
0
int<6>

0
itousize
0

int<6>

0
itousize
0

int<6> USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit

Qubit
Qubit

Qubit

0 1 2
return
0

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

Tag(0)
0

Unit

0
DataflowBlock
0

Input
0 1 2 3

0 1
borrow
0 1

borrow_array<$0,...

Qubit Qubit
Qubit

0 1 2 3 4

Output

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

0 1
isub
0

int<6>

Const(3)
0

0
LoadConst
0
int<6>

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

0 1
isub
0

int<6>

Const(2)
0

0
LoadConst
0
int<6>

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

0 1
borrow
0 1

borrow_array<$0,...

Qubit

load_nat

0

0
ifromusize
0

USize

0

iu_to_s

0

int<6>

0 1
isub
0

int<6>

Const(1)
0

0
LoadConst
0
int<6>

0
itousize
0

int<6>

0
itousize
0

int<6>

USize

Qubit

0 1 2
return
0

borrow_array<$0,...

Qubit
Qubit

Qubit

Qubit

0 1 2
return
0

Qubit

0 1 2
return
0

Qubit

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

USize

borrow_array<$0,...

MakeTuple
0

MakeTuple
0

Tag(0)
0
Unit

Input
0 1 2 3

0 1 2 3 4

Output

Qubit borrow_array<$0,...

Qubit Qubit

Tag(0)
0
Unit

Input
0 1 2 3

0 1 2 3

Output

borrow_array<$0,...

Qubit
Qubit

0
QFree

Qubit

MakeTuple
0

Tag(0)
0
Unit

borrow_array<$0,...Qubit

Qubit

FuncDefn(__main__...)
unitary: 0

0

Input
0 1 2 3

0 1 2 3
CFG
0 1 2 3

Qubit

Qubit Qubit
Qubit

0 1 2 3

Output

Input
0 1 2 3

0
H
0

Qubit

0 1
CX
0 1

Qubit

0 1
CX
0 1

Qubit

0 1
CX
0 1

Qubit

0 1 2 3 4

Output

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

0

Tdg
0

Qubit

0 1
CX
0 1

Qubit

MakeTuple
0

0
H
0

Qubit

MakeTuple
0

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

0

Tdg
0

Qubit

0 1
CX
0 1

Qubit

MakeTuple
0

Qubit

MakeTuple
0

0
T
0

Qubit

0 1
CX
0 1

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0

Tdg
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0
H
0

Qubit

MakeTuple
0

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0

Tdg
0

Qubit

MakeTuple
0

0
H
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Tag(0)
0
Unit

DataflowBlock
0

0
ExitBlock

Qubit

Qubit
Qubit

Qubit

FuncDefn(__main__...)
unitary: 0

0

Input
0 1 2

0 1 2
CFG
0 1 2

Qubit
Qubit

Qubit

0 1 2

Output

Input
0 1 2

0
H
0

Qubit

0 1
CX
0 1

Qubit

0 1
CX
0 1

Qubit

0 1 2 3

Output

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

0

Tdg
0

Qubit

0 1
CX
0 1

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0
T
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Qubit

0

Tdg
0

Qubit

MakeTuple
0

0
H
0

Qubit

MakeTuple
0

Qubit

MakeTuple
0

Tag(0)
0
Unit

DataflowBlock
0

0
ExitBlock

Qubit
Qubit

Qubit

FuncDefn(__main__._rtof)
unitary: 0

0

0 1 2 3
Call
0 1 2

Input
0 1 2

0 1 2
CFG
0 1 2

Qubit
Qubit

Qubit

0 1 2

Output

Input
0 1 2

Qubit
Qubit

Qubit

0 1 2 3

Output

Qubit
Qubit

Qubit

MakeTuple
0

Tag(0)
0

Unit

DataflowBlock
0

0
ExitBlock

Qubit
Qubit

Qubit

FuncDefn(__main__...)
unitary: 0

0

Module

name: __main__
core.used_extensions: [...]
core.generator: {'name':...}